

EE3621: Digital Signal Processing

Lecture 18: Wavelet Transform & Multiresolution Analysis (III B.Tech EEE)

Lecture Notes (40-Minute Plan)

1 Lecture Plan (40 Minutes Breakdown)

- **00:00 – 07:00 (7 mins): Why Wavelets?** Limitations of STFT, time-frequency resolution trade-offs, and constant-Q analysis.
 - **07:00 – 14:00 (7 mins): Continuous Wavelet Transform (CWT):** Definition, mother wavelet, scaling and translation, admissibility condition.
 - **14:00 – 18:00 (4 mins): Common Mother Wavelets:** Haar, Morlet, Mexican Hat, Daubechies.
 - **18:00 – 25:00 (7 mins): Multiresolution Analysis (MRA):** Nested subspaces, scaling function, wavelet function, two-scale relations.
 - **25:00 – 32:00 (7 mins): Discrete Wavelet Transform (DWT):** Filter banks, Mallat's Algorithm, perfect reconstruction.
 - **32:00 – 37:00 (5 mins): Numerical Example:** Haar DWT of a 4-point signal.
 - **37:00 – 40:00 (3 mins): Applications & Checkpoints:** JPEG 2000, ECG, Q&A.
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2 Why Wavelets? The Limitations of STFT

The Fourier Transform provides frequency content but loses time information. The Short-Time Fourier Transform (STFT) addresses this by multiplying the signal $x(t)$ with a sliding window $w(t)$:

$$X(t, \omega) = \int_{-\infty}^{\infty} x(\tau)w(\tau - t)e^{-j\omega\tau}d\tau \quad (1)$$

Physical Intuition: A narrow window provides excellent time resolution but poor frequency resolution. A wide window provides excellent frequency resolution but poor time resolution. In STFT, the window size is fixed, so resolution is the same at all frequencies. Wavelets solve this using **constant-Q analysis**: narrow windows at high frequencies (for transients) and wide windows at low frequencies (for slow drifts).

3 Continuous Wavelet Transform (CWT)

The CWT is defined as:

$$W_x(a, b) = \frac{1}{\sqrt{|a|}} \int_{-\infty}^{\infty} x(t)\psi^*\left(\frac{t-b}{a}\right)dt \quad (2)$$

Where $\psi(t)$ is the mother wavelet, $a > 0$ is the scale, and b is the translation.

For the inverse transform to exist, $\psi(t)$ must satisfy the **admissibility condition**:

$$C_\psi = \int_0^\infty \frac{|\Psi(\omega)|^2}{\omega} d\omega < \infty \quad (3)$$

Theorem: Zero Mean Property

If $C_\psi < \infty$ and $\Psi(\omega)$ is continuous at $\omega = 0$, then $\Psi(0) = 0$.

Proof: Assume $\Psi(0) = c \neq 0$. As $\omega \rightarrow 0$, $|\Psi(\omega)|^2 \rightarrow |c|^2$. The integrand becomes $\frac{|c|^2}{\omega}$, and $\int_0^\epsilon \frac{|c|^2}{\omega} d\omega \rightarrow \infty$, violating $C_\psi < \infty$. Thus, $\Psi(0) = 0$.

4 Common Mother Wavelets

- **Haar Wavelet:** Simplest, a step function. Good for transitions.
- **Morlet Wavelet:** $\psi(t) = e^{-t^2/2} e^{j\omega_0 t}$. Modulated Gaussian, optimal joint time-frequency resolution.
- **Mexican Hat:** $\psi(t) = -\frac{d^2}{dt^2} e^{-t^2/2}$. Second derivative of a Gaussian, used in edge detection.
- **Daubechies:** Compactly supported orthogonal wavelets, used in compression.

5 Multiresolution Analysis (MRA)

MRA defines nested subspaces V_j :

$$\dots \subset V_2 \subset V_1 \subset V_0 \subset V_{-1} \subset \dots \quad (4)$$

A scaling function $\phi(t)$ spans V_0 . The wavelet function $\psi(t)$ spans the orthogonal complement W_0 , where $V_{-1} = V_0 \oplus W_0$.

This leads to the **two-scale relations**:

$$\phi(t) = \sqrt{2} \sum_k h_0[k] \phi(2t - k) \quad (5)$$

$$\psi(t) = \sqrt{2} \sum_k h_1[k] \phi(2t - k) \quad (6)$$

6 Discrete Wavelet Transform (DWT)

The DWT uses a filter bank. The signal is passed through lowpass H_0 and highpass H_1 , then downsampled by 2 to yield approximation cA and detail cD . The Quadrature Mirror Filter (QMF) relation is:

$$h_1[k] = (-1)^k h_0[N - 1 - k] \quad (7)$$

Mallat's algorithm iteratively applies this to cA . A J -level DWT yields $J + 1$ subbands.

6.1 Perfect Reconstruction

The synthesis filter bank conditions for Perfect Reconstruction (PR) are:

$$G_0(z)H_0(-z) + G_1(z)H_1(-z) = 0 \quad (\text{Alias Cancellation}) \quad (8)$$

$$G_0(z)H_0(z) + G_1(z)H_1(z) = 2z^{-L} \quad (\text{Distortion-free}) \quad (9)$$

Cascade (Serial) IIR realization (SOS 1 followed by SOS 2)

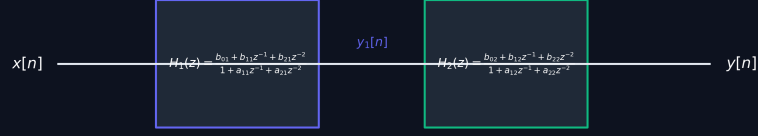


Figure 1: Iterated cascade structure loosely analogous to DWT cascading filter stages.

7 Numerical Example: Haar DWT

1-level Haar DWT of $x = [4, 6, 8, 2]$ using $h_0 = \frac{1}{\sqrt{2}}[1, 1], h_1 = \frac{1}{\sqrt{2}}[1, -1]$.

Approximation cA :

$$\begin{aligned} cA[0] &= 4 \left(\frac{1}{\sqrt{2}} \right) + 6 \left(\frac{1}{\sqrt{2}} \right) = 5\sqrt{2} \\ cA[1] &= 8 \left(\frac{1}{\sqrt{2}} \right) + 2 \left(\frac{1}{\sqrt{2}} \right) = 5\sqrt{2} \end{aligned}$$

Detail cD :

$$\begin{aligned} cD[0] &= 4 \left(\frac{1}{\sqrt{2}} \right) + 6 \left(-\frac{1}{\sqrt{2}} \right) = -\sqrt{2} \\ cD[1] &= 8 \left(\frac{1}{\sqrt{2}} \right) + 2 \left(-\frac{1}{\sqrt{2}} \right) = 3\sqrt{2} \end{aligned}$$

8 Applications

- **Image Compression:** JPEG 2000 uses DWT avoiding blocking artifacts.
- **ECG Denoising:** Thresholding DWT details to remove noise.
- **Seismic & Edge Detection.**

9 Checkpoints

1. **Why STFT struggles?** Fixed window width causes a rigid time-frequency trade-off. Wavelets solve this with variable windows.
2. **Proof of zero mean:** If $\text{mean} \neq 0, \Psi(0) \neq 0$. The admissibility integral $\int \frac{|\Psi|^2}{\omega}$ would diverge at $\omega = 0$.
3. **3-level DWT subbands:** Yields 4 subbands: cD_1, cD_2, cD_3, cA_3 .